

Productive externalities and endogenous growth

Although the Solow models studied so far are quite successful in accounting for many important aspects of economic growth, they have one major limitation: by treating the rate of technological change as exogenous, they leave the economy's long-run (steady state) growth rate unexplained. Solow models therefore belong to the class of so-called *exogenous growth* theories.

In this part of the book we will try to answer a very big question that remains unanswered: how can we explain the rate of technological change which is the source of long-run growth in income per capita? The search for an answer to this fundamental question takes us to the modern theory of *endogenous growth* where the long-run rate of growth in GDP per person is truly endogenous.

A model that 'explains' the long-run rate of growth in GDP per worker is one that endogenizes the underlying rate of technical change, that is, makes this rate depend on basic model parameters. Hence, by an endogenous growth model we mean a model in which the long-run growth rate of technology depends on basic model parameters such as the investment rates in physical and human capital, the population growth rate, or other fundamental characteristics of the economy. An endogenous growth model therefore allows an analysis of how economic policies that affect these basic parameters will affect long-run growth in income per capita. This is an important, and some will say the defining, feature of endogenous growth models: structural economic policy has implications for growth in output per capita in the long run.

In this chapter and the following one we will study endogenous growth theory. The models to be presented can be divided into two categories. In both categories there will be aggregate production functions involving a variable A_t that describes 'technology', but there will be no assumption of exogenous technological progress such as $A_{t+1} = (1 + g)A_t$, where g is exogenous.

One category contains models that include an explicit description of how technological progress, $A_{t+1} - A_t$ in period t , is produced through a specific production process that requires inputs of its own. Since we think of the production of technological progress as arising from research and development, we call such models *R&D-based models of endogenous growth*. These are the subject of the next chapter.

The other category does not have an explicit production process for technological improvement, but assumes that the A_t of the individual firm depends positively on the aggregate use of capital, or of output, because of so-called 'productive externalities'. This implies that the aggregate production function, as opposed to that of the individual firm, will have increasing returns to scale. As we will see, this will result in growth in GDP per worker in the long run without any exogenous technological progress being assumed.

The models in this category are referred to as *endogenous growth models based on productive externalities*, and they are the subject of this chapter.

8.1 A growth model with productive externalities

In Chapter 3 we explained why growth in income per worker had to vanish in the long run according to the basic Solow model. The explanation was related to *constant returns to capital and labour* and the associated *diminishing returns* to capital alone. Let us take the explanation once more in a way that is well suited to our present purposes.

The production function was $Y_t = K_t^\alpha L_t^{1-\alpha}$, exhibiting constant returns to K_t and L_t and diminishing returns to K_t alone ($\alpha < 1$). Consequently there were also diminishing returns to capital per worker in the production of output per worker: $y_t = k_t^\alpha$. Now, assume that there is some growth in the labour force, say at 1 per cent per year. If capital also increases by 1 per cent per year, as in the steady state of the basic Solow model, then because of constant returns to capital and labour, output will increase by 1 per cent per year. Hence output per worker will be constant.

How could there possibly be growth in output per worker? With the production function of the basic Solow model *only* if capital increases by more than 1 per cent per year. If capital increases at a given constant rate of more than 1 per cent per year, say at 2 per cent, then each year output will also increase by more than 1 per cent and hence there will be growth in output per worker. Indeed, the (approximate) growth rate of capital per worker, g_t^k , will be constant and equal to 1 per cent, and the (approximate) growth rate in output per worker, g_t^y , will be $g_t^y = \alpha g_t^k$. Hence, there will be a constant and positive growth rate in output per worker. However, the formula $g_t^y = \alpha g_t^k$ already reveals the problem. Since $\alpha < 1$, the growth rate of income per worker is smaller than the assumed growth rate of capital per worker. Therefore the continued constant growth rate in capital per worker cannot be sustained by savings. What happens is that as long as capital increases faster than labour there will be more and more capital per worker and this implies, due to diminishing returns, that additional units of capital per worker create less and less additional output per worker, and hence less and less additional savings per worker. As a consequence, growth in capital, and in GDP, per worker will have to cease in the long run.

This reasoning suggests that if there were *increasing* returns to capital and labour, then long-run growth in GDP per worker *would* be possible without exogenous technological progress. If both capital and labour were increasing at a rate of 1 per cent per year, then, simply because of increasing returns, output would be increasing by more than 1 per cent per year. And in this case growth would not have to cease in the long run, since it would be unnecessary to build up more and more capital per worker to sustain growth and hence diminishing returns would not be a problem.¹

Increasing returns to scale at the aggregate level therefore seems to be a potential source of endogenous growth.

Constant returns at the firm level and increasing returns at the aggregate level

One may think that the idea of an endogenous growth model suggested by the above reasoning is simply the basic Solow model with the production function of the representative firm exhibiting increasing returns, that is, with the two exponents on capital and labour summing up to a number greater than 1 rather than exactly 1. For two reasons this is not an idea we will pursue. First, because of the replication argument, we believe that there should be close to constant returns at the firm level to the inputs that can be replicated. Second, the

1. In the models in this chapter that formalize this idea, capital and labour do *not* increase at the same rate in equilibrium. Rather capital increases at a greater rate than labour. The argument here is intended to show that increasing returns can work as a source of long-run growth in income per worker. The hypothetical assumption of equal growth rates in capital and labour is made for the sake of the argument.

idea of competitive markets, involving price-taking behaviour of the individual firms, is not compatible with increasing returns at the firm level. Here is why.

Under constant returns to scale and given input prices, total costs are proportional to total output. The reason is simply that an increase in output requires proportional increases in the inputs. Hence, under constant returns average and marginal cost are constant and both equal to the cost of producing one unit, $\hat{C}$, say. Under increasing returns, an increase in output requires less than proportional increases in the inputs, and therefore average and marginal costs will be decreasing in output. With a Cobb–Douglas production function with the exponents adding up to a number greater than 1, the marginal cost, $\hat{C}(Y_t)$, will be a decreasing function, and $\hat{C}(Y_t)$ will go to 0 as Y_t goes to infinity. If the firm takes the prices of inputs and outputs as given, to maximize profits it should want to produce an infinite amount of output. In other words, profit maximization does not imply well-defined behaviour of the individual firm. Price-taking behaviour and perfect competition are not compatible with increasing returns at the firm level.

There is a way to keep the assumption of constant returns at the firm level and at the same time have increasing returns at the aggregate level. In our model we only have one representative profit-maximizing firm, but this firm represents the aggregate behaviour of many firms each of which is small relative to the whole economy. The firm therefore takes aggregate magnitudes such as GDP or the aggregate use of capital as given, since it is too small to have more than a negligible influence on aggregates. In our model the aggregate use of capital has to be equal to the use of capital in the single representative firm, but we should nevertheless assume that *in making its individual decisions, the representative firm takes the aggregates as given*.

We assume that the *individual* production function of the representative firm (in its role as a small individual firm) is:

$$Y_t = (K_t^d)^\alpha (A_t L_t^d)^{1-\alpha}, \quad 0 < \alpha < 1, \quad (1)$$

where the firm takes the labour productivity variable A_t as given and there are constant returns to the inputs of capital and labour, K_t^d and L_t^d . We assume further that the A_t of the *individual* firm depends positively on the *aggregate* stock of capital K_t as expressed by the constant elasticity function:

$$A_t = K_t^\phi, \quad \phi \geq 0. \quad (2)$$

The special case $\phi = 0$ will bring us back to the basic Solow model of Chapter 3, but we will explain below why it may be reasonable to assume $\phi > 0$.

The individual firm has no influence on aggregate capital and takes A_t as given. The *aggregate* production function results from inserting (2) into (1) and using the facts that clearing of the input markets implies $K_t^d = K_t$ and $L_t^d = L_t$, where L_t is total labour supply in period t :

$$Y_t = K_t^\alpha (K_t^\phi L_t)^{1-\alpha} = K_t^{\alpha+\phi(1-\alpha)} L_t^{1-\alpha}. \quad (3)$$

When $\phi > 0$, the aggregate production function in (3) has increasing returns because the sum of the exponents, $1 + \phi(1 - \alpha)$, is larger than 1. A doubling of both aggregate inputs implies that aggregate output is multiplied by a factor $2^{1+\phi(1-\alpha)}$.

Since the individual firm takes aggregate capital as given, the marginal products entering the ‘marginal product equal to real rental rate’ conditions for optimal input demands should be those that appear when A_t is taken as given. Taking partial derivatives in (1), we therefore have:

$$r_t = \alpha \left(\frac{K_t^d}{A_t L_t^d} \right)^{\alpha-1} \quad \text{and} \quad w_t = (1-\alpha) \left(\frac{K_t^d}{A_t L_t^d} \right)^\alpha A_t, \quad (4)$$

in which we can insert the market-clearing conditions $K_t^d = K_t$ and $L_t^d = L_t$ as well as (2) to arrive at:

$$r_t = \alpha \left(\frac{K_t}{K_t^\phi L_t} \right)^{\alpha-1} \quad \text{and} \quad w_t = (1 - \alpha) \left(\frac{K_t}{K_t^\phi L_t} \right)^\alpha K_t^\phi. \quad (5)$$

You will easily verify from (3) and (5) that $r_t K_t = \alpha Y_t$ and $w_t L_t = (1 - \alpha) Y_t$. Hence, we have achieved what we wanted: we have constant returns at the firm level (so perfect competition can be assumed), we have increasing returns at the aggregate level, and our theory of the functional income distribution still has the nice features of constant income shares and no pure profits, capital's share being α .

For this whole construction the assumption of a productive externality from the aggregate use of capital to labour productivity, Eq. (2), is crucial. What could be the motivations for this?

Empirically, the idea of increasing returns at the aggregate level is not implausible.² Estimates of the sum of our exponents, $1 + \phi(1 - \alpha)$, are systematically greater than 1, pointing to $\phi > 0$. Furthermore, they vary widely across investigations, ranging from just above 1 to levels way above 2. There is perhaps a tendency that most estimates (and the most reliable ones) should be found in the (lower) range, 1.1–1.5. A value for the sum of exponents, $1 + \phi(1 - \alpha)$, around 1.3, say, corresponds (with α around $\frac{1}{3}$) to ϕ being around 0.45, while a value of 1.5 would mean that ϕ should be 0.75. This may give an indication of largest possible plausible values for ϕ . Taking the great uncertainty of the estimations into account, however, one cannot completely exclude larger values for the sum of the exponents, for instance values close to $\frac{5}{3}$ (1.67). The latter corresponds to a ϕ around 1, a possibility that will be of importance below.

For the theoretical motivations the key phrase is *learning by doing*. There may be a bit of hand waving involved here, but the idea is that the use of (additional) capital in an individual firm will have a direct effect on production as expressed by the individual production function (1), but in addition it will have an indirect, positive effect on the capabilities of workers who gain new knowledge by working with the new capital. The benefit from this effect does not only accrue to the firm that increases its capital stock, but to firms in general, because knowledge is a public good circulating freely among workers as workers circulate between firms bringing their acquired knowledge and capabilities with them.

This should explain why the additional capability effect spills over to firms in general and hence should be modelled as an externality. But why should there be a capability effect in the first place? Here the idea is that one way workers get more skilful and sophisticated is through the installation of new capital in the firms, since new machinery is a carrier of new technological knowledge. Thus, working with new capital, learning by doing, makes workers more sophisticated. Taking this idea literally, the capability effect should arise from accumulated gross investment rather than from the stock of capital, but we may let the latter approximate all the gross investment undertaken in the past (in this connection ignoring depreciation).³

The complete model

The above explanations have focused on the crucial and only new feature of the growth model we are constructing. In all other respects the model is just like the basic Solow model. We can therefore write down the complete model without further delay.

2. A survey of empirical estimates of returns to scale in aggregate production functions can be found, for example, in Stephanie Schmitt-Grohé, 'Comparing Four Models of Aggregate Fluctuations due to Self-Fulfilling Expectations', *Journal of Economic Theory*, 72, 1997, pp. 96–147.

3. The important idea of productive externalities due to learning by doing comes from a famous article by the economist and Nobel Prize winner Kenneth J. Arrow, 'The Economic Implications of Learning by Doing', *Review of Economic Studies*, 29, 1962, pp. 153–173. Arrow constructs a model that involves a distinction between old and new capital and in which the idea of a capability effect from the use of new capital is perhaps better placed.

Equation (6) below is the individual production function of the representative firm taking A_t as given, but with the equilibrium conditions of the factor markets, $K_t^d = K_t$ and $L_t^d = L_t$, inserted. Equation (7) states the assumption that labour augmenting productivity, A_t , potentially (if $\phi > 0$) depends on aggregate capital because of learning-by-doing productive externalities. The two last equations, (8) and (9), describe capital accumulation and labour force growth, respectively, and are essentially unchanged from the basic Solow model. The parameters are assumed to fulfil $0 < \alpha < 1$, $\phi \geq 0$, $0 < s < 1$, $0 < \delta < 1$, and $n > -1$.

THE COMPLETE GROWTH MODEL BASED ON PRODUCTIVE EXTERNALITIES



$$Y_t = (K_t)^\alpha (A_t L_t)^{1-\alpha}, \quad (6)$$

$$A_t = K_t^\phi, \quad (7)$$

$$K_{t+1} = sY_t + (1-\delta)K_t, \quad K_0 \text{ given} \quad (8)$$

$$L_{t+1} = (1+n)L_t, \quad L_0 \text{ given} \quad (9)$$

We have chosen not to restate the expressions for the real factor prices this time, but whenever needed they can be taken from (5) above. From given initial values, K_0 and L_0 , of the state variables, the model will determine the paths of all the endogenous variables Y_t , K_t , L_t and A_t , and hence, using (5), of the real factor prices.

Sometimes it is assumed in similar models that the external learning-by-doing effect on labour productivity really arises from total production rather than from capital use. This would amount to the formulation, $A_t = Y_t^\phi$, rather than (7) above, the remaining model being the same. An exercise will ask you to verify that this alternative model works qualitatively exactly as the one above. We could also have chosen to let the external effect from capital use (or production) affect total factor productivity rather than the labour-augmenting productivity variable, A_t , without any effect on our conclusions.

8.2 Semi-endogenous growth

As mentioned, in the special case of $\phi = 0$, the model above is just the basic Solow model. We will therefore assume $\phi > 0$, and the aggregate production function (3) will then have increasing returns to K_t and L_t . Note from (3) that if $\phi < 1$, there will be diminishing returns to capital alone, since $\alpha + \phi(1-\alpha)$ will be smaller than 1, while if $\phi = 1$, the aggregate production function will have constant returns to K_t alone. Whether there are diminishing returns or constant returns to capital alone turns out to make an important difference. We will first consider the case $\phi < 1$.

The law of motion

Assuming $\phi < 1$, we can again analyse the model in terms of the technology-adjusted variables, $\tilde{K}_t \equiv k_t/A_t \equiv K_t/(A_t L_t)$ and $\tilde{Y}_t \equiv y_t/A_t \equiv Y_t/(A_t L_t)$, but A_t is no longer growing at an exogenous rate. Instead, the evolution of A_t is endogenous and depends, through (2), on how aggregate capital evolves.

It follows straightforwardly from the production function (6) that:

$$\tilde{Y}_t = \tilde{K}_t^\alpha, \quad (10)$$

and (7) implies that:

$$\frac{A_{t+1}}{A_t} = \left(\frac{K_{t+1}}{K_t} \right)^\phi. \quad (11)$$

These two equations will be used in the following. From the definition of $\tilde{k}_t$ we have:

$$\frac{\tilde{k}_{t+1}}{\tilde{k}_t} = \frac{\frac{K_{t+1}}{K_t}}{\frac{A_{t+1} L_{t+1}}{A_t L_t}} = \frac{\frac{K_{t+1}}{K_t}}{\left(\frac{K_{t+1}}{K_t} \right)^\phi \frac{L_{t+1}}{L_t}} = \frac{1}{1+n} \left(\frac{K_{t+1}}{K_t} \right)^{1-\phi}, \quad (12)$$

where we have used (11) and (9). Inserting that $K_{t+1} = sY_t + (1-\delta)K_t$, we get:

$$\begin{aligned} \frac{\tilde{k}_{t+1}}{\tilde{k}_t} &= \frac{1}{1+n} \left(\frac{sY_t + (1-\delta)K_t}{K_t} \right)^{1-\phi} \\ &= \frac{1}{1+n} \left(\frac{s\tilde{y}_t}{\tilde{k}_t} + (1-\delta) \right)^{1-\phi} \\ &= \frac{1}{1+n} [s\tilde{k}_t^{\alpha-1} + (1-\delta)]^{1-\phi}, \end{aligned}$$

where we have used (10) for the latter equality. Multiplying on both sides by $\tilde{k}_t$ gives the transition equation:

$$\tilde{k}_{t+1} = \frac{1}{1+n} \tilde{k}_t [s\tilde{k}_t^{\alpha-1} + (1-\delta)]^{1-\phi}, \quad (13)$$

which we can also write in the form:

$$\tilde{k}_{t+1} = \frac{1}{1+n} \left[s\tilde{k}_t^{\frac{\alpha+\phi-\alpha\phi}{1-\phi}} + (1-\delta)\tilde{k}_t^{\frac{1}{1-\phi}} \right]^{1-\phi}. \quad (14)$$

(You should check this reformulation.) Note that since we have assumed $\phi < 1$, the exponents in (14) are all well defined and positive. Also note that if we set $\phi = 0$, both of the equations (13) and (14) become the transition equation of the basic Solow model (the model of Chapter 3 with $B = 1$).

Convergence to steady state

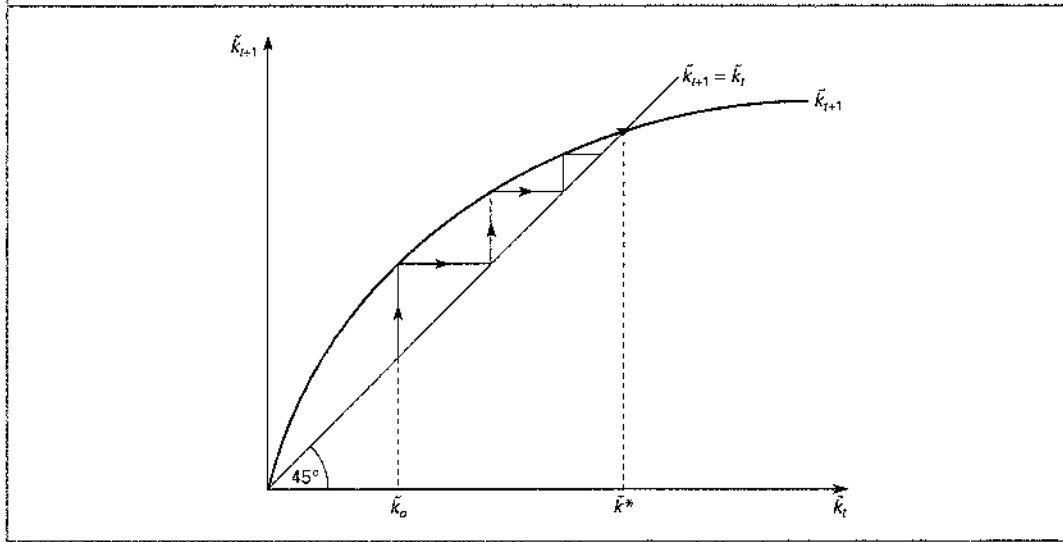
The transition equation has the form of a one-dimensional first-order difference equation in $\tilde{k}_t$. We will establish properties of the transition equation implying that in the long run $\tilde{k}_t$ converges to a specific value $\tilde{k}^*$.

From either of (13) or (14), it is clear that the transition curve passes through $(0, 0)$. From (14) it follows that it is everywhere increasing. Further, from (13) follows that it crosses the 45° line for exactly one positive value $\tilde{k}^*$ of $\tilde{k}_t$; inserting $\tilde{k}_{t+1} = \tilde{k}_t$ into (13) and solving for $\tilde{k}_t$ gives

$$\tilde{k}_t = \tilde{k}^* \equiv \left(\frac{s}{(1+n)^{1/(1-\phi)} - (1-\delta)} \right)^{\frac{1}{1-\alpha}}, \quad (15)$$

verifying a unique positive intersection if $(1+n)^{1/(1-\phi)} > 1-\delta$. The latter is assumed; note that it follows, e.g., from n positive (larger than or equal to zero).

Finally, for convergence to $\tilde{k}^*$ it is important that at $\tilde{k}^*$ the transition curve intersects the 45° line from above, as illustrated in Fig. 8.1. To show this, one can differentiate the transition function given in (13) with respect to $\tilde{k}_t$, insert $\tilde{k}^*$ for $\tilde{k}_t$ to find the slope of the transition curve at $\tilde{k}^*$, and then derive the condition for this slope being less than 1. You should do this and find the condition $(1+n)^{1/(1-\phi)} > 1-\delta$, that we have just assumed.

FIGURE 8.1 The transition diagram of the model of semi-endogenous growth

The properties established and illustrated in Fig. 8.1 imply that in the long run $\tilde{k}_t$ will converge to $\tilde{k}^*$. Consequently, $\tilde{y}_t = \tilde{k}_t^\alpha$ will converge to the associated:

$$\tilde{y}^* \equiv \left(\frac{s}{(1+n)^{1/(1-\phi)} - (1-\delta)} \right)^{\frac{\alpha}{1-\alpha}}. \quad (16)$$

We have demonstrated long-run convergence of $\tilde{k}_t$ and $\tilde{y}_t$ to $\tilde{k}^*$ and $\tilde{y}^*$, respectively. This defines the steady state.

Semi-endogenous growth in steady state

Our conclusions so far seem reminiscent of those we arrived at in Chapter 5 for the general Solow model. The auxiliary variables are defined in the same way, $\tilde{k}_t \equiv k_t/A_t$ and $\tilde{y}_t = y_t/A_t$, respectively, and we have found again that these variables converge to constant steady state values. When they are locked at these constant values in steady state, both k_t and y_t must grow at the same rate as A_t . In the Solow model this sufficed for determining the steady state growth rates of k_t and y_t . Both had to be equal to the exogenous growth rate, g , of A_t . This time we do not have an exogenous growth rate of A_t , so to determine the steady state growth rate of y_t we must determine the *endogenous* growth rate of A_t in steady state.

This is easy to do. Consider (12) above. In steady state the left-hand side, $\tilde{k}_{t+1}/\tilde{k}_t$, is equal to 1. Therefore the right-hand side must also equal 1:

$$\frac{1}{1+n} \left(\frac{K_{t+1}}{K_t} \right)^{1-\phi} = 1 \Leftrightarrow \frac{K_{t+1}}{K_t} = (1+n)^{\frac{1}{1-\phi}}.$$

Using (11) then gives:

$$\frac{A_{t+1}}{A_t} = (1+n)^{\frac{\phi}{1-\phi}} \Leftrightarrow \frac{A_{t+1} - A_t}{A_t} = (1+n)^{\frac{\phi}{1-\phi}} - 1 \equiv g_{se} \quad (17)$$

Hence, in steady state the growth rate of A_t is $(1+n)^{\phi/(1-\phi)} - 1$, and we call this growth rate g_{se} . (The constant approximate growth rate, $g_{se}^A \equiv \ln A_{t+1} - \ln A_t$, is $g_{se}^A \approx \phi n/(1-\phi)$, as you should verify.) Note that g_{se} is determined endogenously and depends on model parameters.

In steady state the growth rates of capital per worker, k_t , and GDP per worker, y_t , must then also be g_{se} . Hence from (17), if the growth rate of the labour force is 0, then the steady state growth rate of GDP per capita is also 0. For the growth rate of GDP per capita to be positive, a positive population (labour force) growth rate is required. This should not be surprising in view of the remarks in Section 8.1. To exploit the increasing returns in the aggregate production function an increasing labour force is required. The term 'semi-endogenous' growth refers to the fact that we only have (endogenous) growth in GDP per worker if there is (exogenous) population growth.

SEMI-ENDOGENOUS GROWTH



Under the highly plausible stability condition, $(1+n)^{1/(1-\phi)} > 1-\delta$, and under the assumption that the elasticity from aggregate capital use to labour-augmenting productivity is smaller than 1, the growth model based on productive externalities implies that capital per effective worker, k_t/A_t , and income per effective worker, y_t/A_t , converge to certain steady state values. Hence in the long run capital per worker, k_t , and income per worker, y_t , must both converge on growing at the same rate as does A_t , implying that the capital–output ratio, $k_t/y_t = K_t/Y_t$, converges to a constant value in accordance with balanced growth. In steady state, the common growth rate, g_{se} , of A_t and y_t depends critically on the growth rate, n , of the labour force. In particular, g_{se} is strictly positive if and only if n is strictly positive.

Note that in steady state the inputs of capital and labour are not growing at the same rates. Labour input grows at rate n , but since $k_t \equiv K_t/L_t$ grows at the rate g_{se} , capital must be growing at approximately the rate $n + g_{se}$. Only if $n = 0$ are these growth rates equal.

Is the endogenous growth rate g_{se} of GDP per person derived from this model of a realistic size for plausible parameter values? Annual population growth rates of around 0.5 per cent are typical for Western countries (see Table A at the end of Book One) and according to the empirical studies mentioned earlier, realistic values of ϕ could be up to 0.5–0.75. As you will easily verify from (17), these parameter values result in steady state growth rates, g_{se} , in the range from 0.5 per cent to 1.5 per cent, which is not far-fetched.

The steady state growth path and structural policy

Let us now turn to the steady state growth path. From the definition of $\tilde{y}_t$, the steady state growth path of y_t must be $y_t^* = \tilde{y}^* A_t$. The evolution of A_t obeys $A_t = K_t^\phi$, so $y_t^* = \tilde{y}^* K_t^\phi$. Furthermore, when the economy is in steady state there is a necessary link between K_t and L_t . This follows because in steady state the variable $\tilde{K}_t$, defined as $K_t/(A_t L_t) = K_t^{1-\phi}/L_t$, is locked at the value $\tilde{K}^*$, so in steady state:

$$K_t = (\tilde{K}^* L_t)^{\frac{1}{1-\phi}}.$$

Inserting this expression for K_t into the steady state growth path, $y_t^* = \tilde{y}^* K_t^\phi$, for output per worker gives:

$$y_t^* = \tilde{y}^* (\tilde{K}^* L_t)^{\frac{\phi}{1-\phi}} = (\tilde{K}^*)^{\frac{\phi}{1-\phi}} L_0^{\frac{\phi}{1-\phi}} (1+n)^{\frac{\phi}{1-\phi} t},$$

where we have used that $\tilde{y}_t = \tilde{K}_t^\alpha$, and $L_t = (1+n)^t L_0$. We can now finally insert $\tilde{K}^*$ from (15), and from (17) that $(1+n)^{\frac{\phi}{1-\phi}} = 1 + g_{se}$:

$$y_t^* = \left(\frac{s}{(1+n)^{\frac{1}{1-\phi}} - (1-\delta)} \right)^{\left(\frac{\alpha + \frac{\phi}{1-\phi}}{1-\phi} \right) / (1-\alpha)} L_0^{\frac{\phi}{1-\phi}} (1 + g_{se})^t. \quad (18)$$

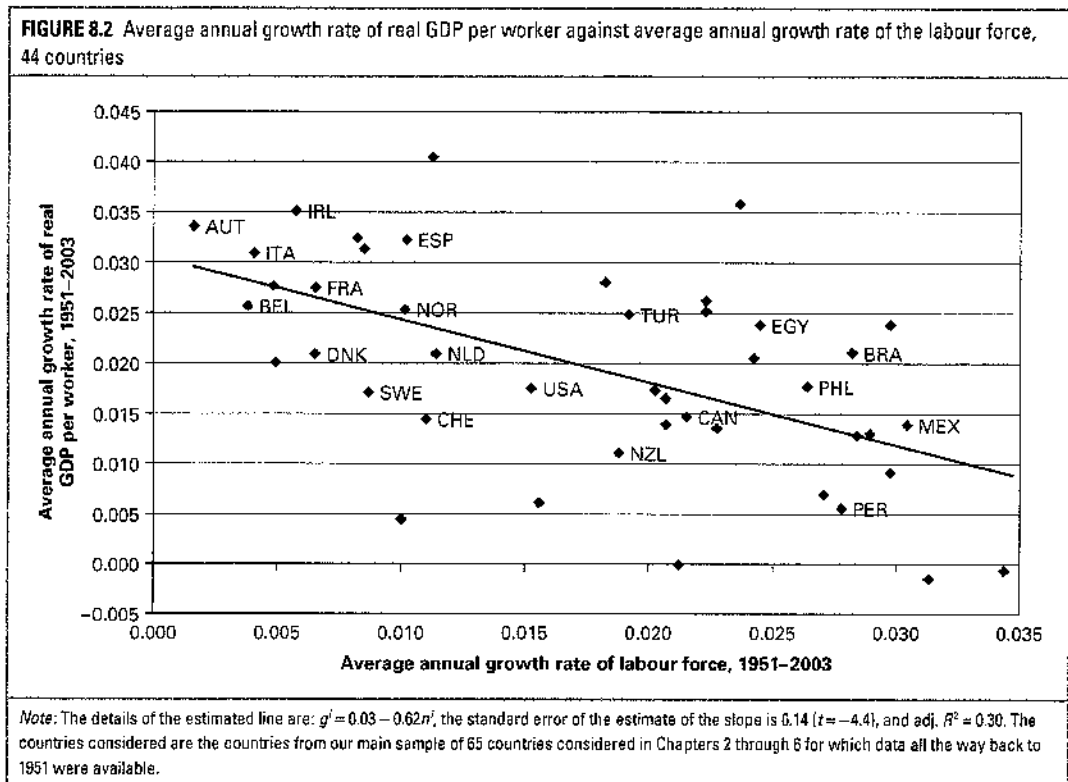
This gives the steady state growth path (y_t^*) for output per worker. The corresponding steady state growth path (c_t^*) for consumption per worker is obtained by multiplying by $1 - s$ on both sides of (18).

Structural economic policies for steady state must work to affect the positions of these paths or the growth rates along them (or both). With respect to the positions, the policy implications are somewhat similar to those in the basic Solow model. For example, a higher investment rate, s , shifts the growth path for y_t upwards. Note, however, that the golden rule value for s , implying the highest possible position of (c_t), is now $\alpha + \phi(1 - \alpha)$, which is greater than the α we have arrived at for the basic Solow model. (As an exercise, show that the golden rule saving rate is $\alpha + \phi(1 - \alpha)$ and try to explain intuitively why for a given α we must now get a higher golden rule savings rate.)

The really new feature is that the steady state growth rates of y_t and c_t depend endogenously on model parameters: both are equal to $g_{ss} \equiv (1 + n)^{\phi/(1-\phi)} - 1$. Taking ϕ to be a given technical parameter (not easily affected by policy), the conclusion from this model's steady state is that to promote long-run growth in GDP and consumption per capita, one should try to *promote* population growth! There are reasons why one would be cautious about such a policy.

Empirics for semi-endogenous growth

The model's steady state predicts that a higher growth rate in the labour force, or the population, should give higher growth in GDP per capita. One way to test this prediction is to plot average annual growth rates in GDP per worker over some period against average annual labour force growth rates over the same period across countries, just as we did in Fig. 7.3, finding a negative, rather than a positive, relation. Figure 8.2 does more or less the same, focusing on a longer period and fewer countries.



We find (again) a rather clear *negative* relationship between labour force growth and economic growth across countries in the period considered. The OLS-estimated straight line has a significantly negative slope (the details of the estimation are reported in the figure legend). This does not necessarily mean that the model of semi-endogenous growth is wrong, however.

First, it is not clear exactly what kind of area the model covers, a region, a country, or perhaps the (developed) world, since it is not easy to tell how wide the crucial external, spillover effect of the model reaches. If the firms and workers in one country learn from experiences with production and capital use in other countries, then perhaps the model should be considered to cover the world. It is therefore not clear if cross-country evidence as reported in Fig. 8.2 is relevant.

Second, the model we consider has a steady state *and* it has convergence to the steady state in the long run. During the convergence period there will be transitory growth in addition to the underlying (endogenous) steady state growth, and the transitory growth depends negatively on population growth, just as in the Solow models, and for the same reason: a decrease in the rate of population growth shifts the steady state growth path up but reduces its slope (see Eq. (18)). Hence, during the transition to the new steady state growth path there will, for some time, be a positive transitory contribution to growth arising from the lower population growth. Furthermore, convergence in the model we consider may be quite slow due to the presence of the productive externality. As ϕ comes close to 1, convergence will be very slow. (Exercises 1 and 2 at the end of the chapter will ask you to look in more detail at the convergence process, both numerically and analytically.) What we have found in Fig. 8.2 may be interpreted as an expression of long-lasting transitory growth that should indeed depend negatively on population growth according to our model.

These remarks seem to call for (even) longer-run empirical evidence that does not go across countries. Table 8.1 reports for each of the two sub-periods 1870–1930 and 1930–90

TABLE 8.1 Average annual growth rates in population and real GDP per capita in 17 industrialized countries, 1870–1990

	Average annual population growth rate (%)		Average annual growth rate of GDP per capita (%)	
	1870–1930	1930–90	1870–1930	1930–90
Australia	2.3	1.6	0.4	2.0
Austria	0.7	0.2	1.1	2.6
Belgium	0.8	0.4	1.0	2.1
Canada	1.7	1.6	1.7	2.4
Denmark	1.0	0.6	1.6	2.1
Finland	1.1	0.6	1.4	3.1
France	0.1	0.5	1.5	2.3
Germany	1.0	0.7	1.2	2.5
Italy	0.6	0.6	1.1	2.9
Japan	1.0	1.1	1.5	3.9
Netherlands	1.3	1.1	1.2	1.8
New Zealand	2.7	1.4	1.2	1.6
Norway	0.8	0.7	1.6	2.7
Sweden	0.6	0.6	1.4	2.5
Switzerland	0.7	0.8	1.7	2.1
UK	0.7	0.4	0.8	1.9
USA	1.9	1.2	1.5	2.1

Source: Angus Maddison, *Monitoring the World Economy 1820–1992*, Paris, OECD, 1995.

the average annual population growth rate and the average annual growth rate of GDP per capita for a number of countries. For all of them economic growth increased substantially from the early sub-period to the later one, while for all countries but three, population growth decreased, for many of the countries substantially. So, also if we consider quite long periods and 'stay' within each country or within the group of all countries, the evidence shows little sign that growth in GDP per capita should be positively related to population growth.⁴

EMPIRICS AND POLICY CONCERNING STEADY STATE



The position of the long-run, balanced growth path of the model of semi-endogenous growth depends on structural parameters in a way that supports the same kind of policy conclusions as were derived from the Solow model of exogenous growth: high investment rates and low labour force growth rates will elevate the growth path to a high position, now with a golden rule savings rate that exceeds capital's share, reflecting the positive spillover effect from capital use. However, the way that the growth rate along the balanced growth path depends on parameters suggests that promoting population growth will promote economic growth. The prediction of a positive association between population growth and economic growth is not easy to reconcile with the empirics.

We can still not conclude from the empirics above that the model of semi-endogenous growth is wrong. It could be that convergence according to the model is so slow that transitory growth really takes place over periods as long as 120 years. In that case the general picture found in Table 8.1 could be compatible with the model. However, if convergence to steady state is that slow, then the model's steady state is not very descriptive for periods of interest, so we should consider the very slow convergence process itself as the model's main prediction. This is exactly the perspective taken in the next section. We will apply a specific assumption that makes the convergence period not just very long, but infinite. We will then consider the model's infinite, outside steady state behaviour as an approximation of a very long-lasting convergence process.

8.3 Endogenous growth

In Exercise 2 (which is an important exercise) you are asked to find an analytical expression for the rate of convergence for y^* according to our model with $\phi < 1$. For the case where population growth is small, $n \approx 0$, the rate is $\lambda \approx (1 - \alpha)(1 - \phi)\delta$. This verifies the claim above that a large ϕ less than 1 implies very slow convergence. Below we will indeed set $n = 0$, since we now want to investigate the possibility that over periods of interest, such as centuries, population growth is *not* the factor that causes economic growth. The size of the labour force is then constant, equal to L , say. The idea is to consider the growth model of this chapter with $\phi = 1$. This gives a zero rate of convergence and hence an everlasting convergence process. We consider the everlasting convergence resulting from $\phi = 1$ to be of interest because it approximates the very long-lasting convergence that would result from a ϕ that is large, but still less than unity.⁵

4. It should be noted that Table 8.1 states population, rather than labour force, growth rates, and growth rates of GDP per capita rather than of GDP per worker, so it does not take changes in the evolution of participation rates into account.

5. It is *only* as an approximation of the case of a large ϕ less than one that $\phi = 1$ is of interest. There is no reason to believe that a parameter value should take one particular knife-edge value.

The AK model

In the analysis in the previous section the term $1 - \phi$ appeared in many places, including in some denominators, and several expressions would be meaningless for $\phi = 1$. Hence we cannot simply set $\phi = 1$ in all the equations above. For what we did in Section 8.1, on the other hand, it was not important whether ϕ was smaller than or equal to 1, so the expressions for the real factor prices we found there and the associated theory of the income distribution are also relevant when $\phi = 1$. With $L_t = L$ and $\phi = 1$ we get from (5) that:

$$r_t = r \equiv \alpha L^{1-\alpha} \quad \text{and} \quad w_t = (1 - \alpha)K_t/L^\alpha, \quad (19)$$

so the real interest rate, $r - \delta$, is constant and the wage rate, w_t , evolves proportionally to K_t , and hence to $k_t \equiv K_t/L$.

In addition to the above expressions for the real factor prices the model can be condensed to the two equations:

$$Y_t = K_t L^{1-\alpha} \equiv AK_t, \quad (20)$$

$$K_{t+1} = sY_t + (1 - \delta)K_t, \quad (21)$$

where the first equation is the aggregate production function resulting from (6) and (7), or directly from (3), with $\phi = 1$, while the second equation is the usual capital accumulation equation repeated from (8). The model's last equation, (9), has been replaced by $L_t = L$. Note that in (20) we have used the definition $A \equiv L^{1-\alpha}$. This is a bit of an abuse of notation, but it should cause no confusion if we denote $L^{1-\alpha}$ by an A without a subscript t . We apply this notation because the model we consider is so often referred to as the 'AK model'.

Growth according to the AK model

It is easy to see that the model above can result in permanent growth in GDP per worker. Dividing both sides of the production function (20) by L , in order to transform variables into per worker terms, gives $y_t = Ak_t$. Dividing also both sides of the capital accumulation equation (21) by L , and inserting Ak_t for y_t , then gives the transition equation:

$$k_{t+1} = (sA + 1 - \delta)k_t, \quad (22)$$

Subtracting k_t from both sides gives the Solow equation:

$$k_{t+1} - k_t = sAk_t - \delta k_t, \quad (23)$$

and then dividing both sides by k_t gives the modified Solow equation:

$$\frac{k_{t+1} - k_t}{k_t} = sA - \delta \equiv g_e. \quad (24)$$

The latter equation directly gives the constant and endogenous growth rate, g_e , in capital per worker, k_t , and hence in capital, K_t , since $k_t = K_t/L$. Furthermore, since y_t is a constant times k_t , the growth rate of output per worker is g_e , and hence the growth rate of consumption per worker, $c_t = (1 - s)y_t$, must be g_e . Finally, the technology variable A_t (not to be confused with the constant $A \equiv L^{1-\alpha}$) must increase at rate g_e , since with $\phi = 1$, we have $A_t = K_t$.

It follows that according to the AK model, $g_e = sA - \delta$ is the common endogenous growth rate of all the variables we are interested in (we assume that $sA - \delta > 0$). Hence, everlasting economic growth is possible in this model without an assumption of exogenous technological growth and without population growth. To understand this result it may be useful to draw some well-known diagrams.

FIGURE 8.3 The transition diagram (left), the Solow diagram (right), and the modified Solow diagram (bottom left) of the model of endogenous growth

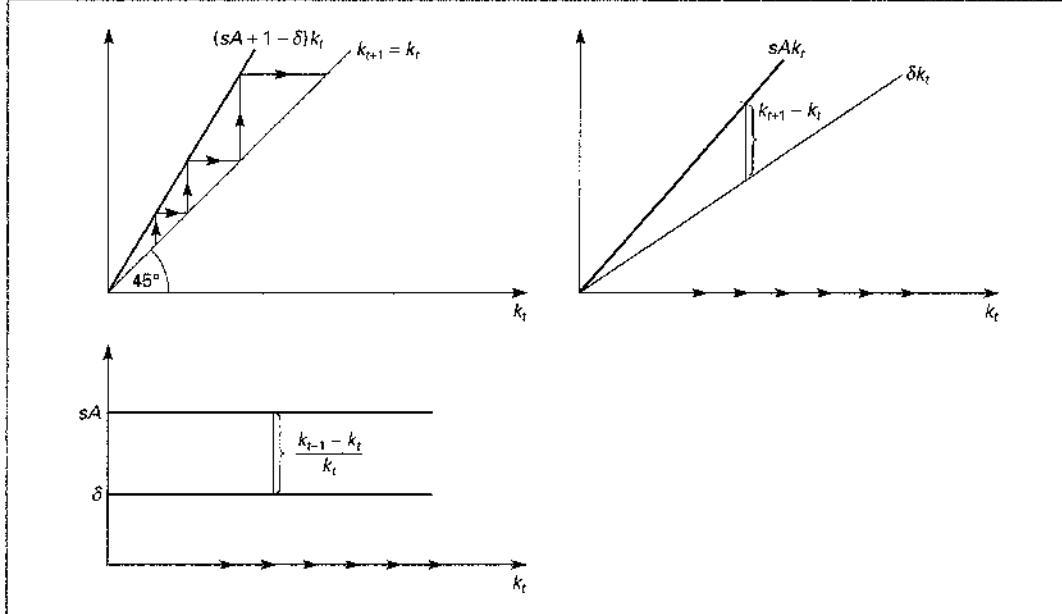


Figure 8.3 shows the transition diagram associated with (22), the Solow diagram associated with (23), and the modified Solow diagram associated with (24). The diagrams illustrate that in this model there is no steady state and k_t (and hence y_t and Y_t) grow at a constant rate forever.

The ‘curves’, which are straight lines in Fig. 8.3, would in Solow models be true curves due to diminishing returns to capital per worker ($\alpha < 1$). The linearity of the curves in the present model reflects that there are no diminishing returns to capital *at the aggregate level*. Rather there are constant returns to capital per worker: $y_t = Ak_t$.

The ‘growth brake’ from the Solow models, diminishing returns to capital, is simply no longer present in the aggregate production function. The source of growth is thus aggregate constant returns to the reproducible factor, capital.

Implications for structural policy and the scale effect

The main conclusion from the model is that a *higher savings (investment) rate, s , gives rise to a permanently higher growth rate in GDP and consumption per worker*. This differs from our earlier conclusions since a higher s no longer just gives a higher *level* of output per worker in the long run and a *temporarily* higher transitory growth rate in the intermediate run, but it results in a *permanently* higher rate of growth in output per worker. Since our model with $\phi = 1$ should be seen as an approximation of the semi-endogenous growth model with a large ϕ smaller than 1, we should remember that strictly speaking the correct statement is that an increase in s gives *very long-lasting* transitory growth in GDP per worker. The implications for economic policy are obvious: policies that stimulate savings now give a very long-lasting boost to growth.

A decrease in δ has an effect similar to that of an increased s , but it may be more difficult to achieve through economic policy. More effective aggregate investment should, however, lead to a lower rate of depreciation, and a lower rate of depreciation leads, in the model of endogenous growth, to a permanently higher growth rate in GDP per worker. If, somehow,

a government can take actions that make net investment more 'effective', it can perhaps take advantage of this effect. This may be most relevant for countries where the government is heavily involved in production so that a large part of investment is not driven by private incentives, but decided upon by government bureaucracies. Historically this method of making investment decisions has often led to inefficient and rapidly worn out (sometimes even useless) investment. By relying more on private, profit-motivated incentives in investment decisions, such a country could probably attain higher prosperity and growth in the long run, although there may be transition costs involved in changing from one type of economic system to another.

Policies for more effective investment are not only relevant in connection with endogenous growth. In the Solow model a lower depreciation rate results in a higher *level* of GDP per capita in the long run, which is also important. Note the general tendency that parameter changes that give a long-run *level* effect on GDP per worker in the Solow models give a long-run effect on the *rate of change* in GDP per worker in the model of endogenous growth.

There is one odd feature of the AK model which casts doubt on any policy recommendation derived from it. Remember that $A \equiv L^{1-\alpha}$. Hence the growth rate $g_a = sA - \delta$ is higher the larger the constant population size L is, and the growth rate would even increase and indeed explode over time if population increased by a constant rate. This is the so-called 'scale effect', which is rather controversial. Empirical evidence does not support a hypothesis that larger countries should have larger long-run growth rates, but again it is not clear what geographical area the model covers. Perhaps it is the world, but then during the past 200 years the world population has increased rapidly while economic growth rates have not shown a similarly strong increase.

It is possible to get rid of the scale effect. Assume that the productive externality, which is the driving force of endogenous growth in this chapter, arises from capital *per worker* rather than from capital itself, so that Eq. (7) in our model should be replaced by:

$$A_t = \left(\frac{K_t}{L_t} \right)^\phi.$$

You will easily verify that with $\phi = 1$ the aggregate production function becomes $Y_t = K_t$, that is, the A (without subscript t) is equal to 1 and independent of L . Consequently the endogenous growth rate of the model will be $g_c = s - \delta$, and the scale effect has been eliminated. However, aggregate production (not production in the individual firm) is now completely insensitive to labour input. The positive effect on production of a higher labour input, L_t , arising at the firm level is exactly offset by the negative external effect of a lower K_t/L_t at the aggregate level (when $\phi = 1$).

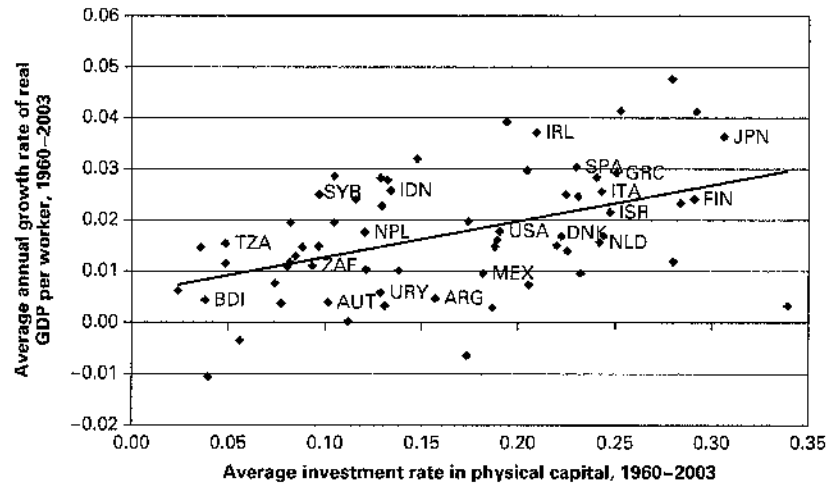
Either endogenous growth models have the unattractive scale effect or they assume, unattractively, that labour inputs are unproductive at the aggregate level. There is no way around this problem.

Empirics for endogenous growth

As mentioned, a main prediction of the model of endogenous growth is the positive influence of the savings or investment rate on the long-run growth rate of GDP per worker. Let us put this prediction to a test by considering cross-country evidence. Figure 8.4 plots average annual growth rates, g^i , from 1960 to 2000 against average investment rates, s^i , over the same period across countries i . The sample of 65 representative countries is the one considered in earlier chapters as well.

The figure shows a quite good and tight positive relationship. This is a main reason why economists take the theories of endogenous growth seriously (still remembering that these should be seen as approximations of theories of very long-lasting transitory growth).

FIGURE 8.4 Average annual growth rate of real GDP per worker against average investment rate in physical capital, 65 countries



Note: The estimated line is $\hat{g}^i = 0.006 + 0.071s^i$, the standard error of the estimate of the slope is 0.017 ($t = 4.2$), and adj. $R^2 = 0.21$.

Source: Penn World Table 6.2

ENDOGENOUS GROWTH



The growth model based on productive externalities with a unit elasticity from aggregate capital to labour-augmenting productivity gives rise to permanent growth in income per capita even without population growth. A higher savings rate gives rise to a permanently higher growth rate in GDP and consumption per worker. Hence, economic policy that stimulates saving not only has a permanent, positive effect on the level of GDP per capita – it also has a permanent, positive effect on the growth rate of GDP per capita. The model feature of a positive long-run association between the investment rate and the growth rate of GDP per worker is in nice accordance with cross-country empirics. However, the model suffers from an implausible scale effect according to which a larger labour force should result in a permanently larger growth rate of GDP per worker.

Yet the positive empirical relationship does not *prove* that the idea of endogenous growth is correct. As we have argued several times, it is not clear whether the model covers a country or the world, so cross-country evidence may not be the appropriate kind of empirics to study.

Furthermore, in no way does Fig. 8.4 contradict the traditional Solow models according to which an increase in the savings rate will imply a (higher) transitory growth in excess of the underlying exogenous growth. Perhaps the figure just shows that countries with a high average s^i over the period considered typically have experienced increases in s^i and therefore have had relatively high transitory growth rates. However, the next section will provide a reason for indeed viewing Fig. 8.4 as indicating endogenous growth.

8.4 Exogenous versus endogenous growth

In this chapter we have studied two distinct and alternative types of endogenous growth, Isemi-endogenous and endogenous growth, both fundamentally caused by productive externalities.

The presence of a positive, but not too strong productive externality, $0 < \phi < 1$, gave rise to a model of semi-endogenous growth (Section 8.2). This model had convergence to a steady state in the long run, and therefore it had the 'convergence property': growth is faster the further below steady state the economy is. Furthermore, in steady state economic growth was explained by growth in the labour force, and the long-run growth rate of GDP per worker was higher the higher the rate of population growth.

A strong productive externality, $\phi = 1$, resulted in a model of (genuinely) endogenous growth (Section 8.3). This model had no steady state and hence no convergence to one (or rather, in the appropriate model interpretation, very slow convergence to one). Therefore the model did not have the convergence property. Growth in GDP per capita could occur without population growth, but there was a scale effect: a larger labour force gave a higher rate of economic growth, and a fixed population growth would imply an exploding economic growth rate.

Both the model of semi-endogenous growth and the model of endogenous growth delivered explanations of economic growth: *the growth rate of GDP per worker depended on basic structural model parameters.*

In previous chapters we have studied models of exogenous growth. All these models had convergence to a steady state and the convergence property. In the long run the growth rate of GDP per worker was either independent of or negatively affected by the population growth rate (the latter in the realistic presence of natural resources). In these models *permanent growth in GDP per worker was a reflection of exogenous technological growth*, so the models did not really explain long-run economic growth.

The basic mechanism for endogenous growth in this chapter was productive externalities. The technological growth arising from these came as an unintended by-product of economic activity. No agents were concerned with deliberately producing technological progress. In the real world we know that a lot of research and development activities *intended* to create technological progress are undertaken in private companies as well as at universities. Therefore, perhaps the endogenous growth models of the next chapter in which technological progress is the outcome of an explicit and deliberate production activity will be more convincing.

However, the simple externality-based models of endogenous growth considered in this chapter share enough features with the more advanced endogenous growth models to be representative for endogenous growth theory. Therefore we can already present some main arguments from an ongoing and fascinating debate among growth theorists. The issue in this debate is whether the economic growth we see in the real world is best understood by exogenous growth models or by endogenous growth models. Here are some main arguments from the debate.

Explaining growth

Everybody agrees that it is good to have explanations of long-run economic growth, and since the endogenous growth models deliver much more of an explanation than exogenous growth models, this is certainly an argument in favour of endogenous growth models. However, it could be that at the present stage of knowledge the existing endogenous growth models were not convincing, so one would have to rely on exogenous growth models for understanding growth, although the understanding would be limited by technological progress being unexplained. Whether the growth explanations and other features of the endogenous growth models are sufficiently convincing is mainly an empirical matter, and some arguments presented below may be important for deciding this.

The knife-edge argument

Strictly speaking, the model of (genuinely) endogenous growth only pertains to a zero probability, knife-edge case ($\phi = 1$) which is uninteresting. Endogenous growth models are

sometimes criticized on such grounds. However, as we have carefully argued, the model should rightly be interpreted as an approximation of a wider case (ϕ slightly smaller than 1), which does not have zero probability. On the face of it, this criticism is therefore not valid. The issue is really whether the relevant parameter (here ϕ) can realistically be assumed to be so close to a limiting value (here 1) that the model of endogenous growth can be seen as a good approximation. This is again an empirical matter. For the externality parameter ϕ considered in this chapter, empirical evidence seems to point to positive values of at most up to 0.75, which speaks in favour of models of semi-endogenous growth.

Population growth, economic growth and the scale effect

If the knife-edge property of the endogenous growth model does not represent a serious objection to it, the scale effect does. The feature that an increasing population implies an increasing growth rate in GDP per capita is simply not realistic. The presence of the scale effect is the single most important argument against models of (genuinely) endogenous growth.

According to the model of *semi*-endogenous growth there should be a positive relationship between the long-run growth rate of GDP per capita and the (constant) population growth rate. Based on the evidence presented in Fig. 8.2 and Table 8.1, one may have a hard time finding this feature convincing. There is, however, a counterargument: what Table 8.1 shows is true, but another fact is, according to this argument, more impressive and more important in a very long-run perspective. As we have seen, many countries in the West have experienced relatively constant average annual growth rates in GDP per capita of about 2 per cent during the past 200 years. Over the several thousands of years before that, the average annual growth rates in GDP per capita must have been close to 0 to fit with the level of income per capita around year 1800. So, thousands of years of almost no economic growth have been succeeded by 200 years of rapid economic growth. Population growth has behaved basically the same way, with the world population being close to constant for hundreds of years up to the eighteenth century and thereafter increasing rapidly.⁶ This actually fits with semi-endogenous growth. Still, one may wonder if it is really the world's population growth that has caused economic growth, or the other way round. Furthermore, Table 8.1 shows that the areas in the world that have experienced the high (and even increasing) economic growth have had decreasing population growth. It is hard to believe that population growth in Africa, India and China should have caused economic growth in Europe, the USA and (other) parts of Asia, while it seems more plausible that economic growth in Europe and the USA has (indirectly) contributed to reduce mortality and create population growth in Africa and parts of Asia.

In the eyes of the authors of this book, the empirical relationship between population growth and economic growth seems to be mostly in favour of the models of exogenous growth, which predict a negative relationship when natural resources are included in the model.

Convergence

We saw in Chapter 2 that focusing on countries that can be assumed to be structurally similar, such as the OECD countries, there is a clear tendency for the countries that are initially most behind to grow the fastest and thus catch up to the initially richest.⁷ In Chapters 5 and 6 we saw that if one controls for structural differences between countries in a way suggested by exogenous growth models, then also for larger, more representative groups of countries, the initially poorest tend to grow the fastest. There is thus evidence in support of convergence.

In the world of models the convergence among countries arises naturally in models with (an appropriate form of) convergence to a steady state. As the steady state is approached

6. For these 'facts' and an interesting discussion of them, we refer to Michael Kremer, 'Population Growth and Technological Change: One Million B.C. to 1990', *Quarterly Journal of Economics*, 108, 1993, pp. 681–716.

7. This pattern is even more clear between states in the USA.

(from below), growth becomes slower and slower. Two different countries with similar characteristics will converge to the same steady state, so the country starting out with the lowest GDP per person will grow the fastest. The empirically observed convergence between countries is thus most naturally explained by models of exogenous or semi-endogenous growth.

The endogenous growth model considered in Section 8.3, on the other hand, did not have convergence to a steady state and showed no tendency for the initial position to be important for subsequent growth. The observed convergence thus contradicts the model and seems to speak against endogenous growth. It must be added, however, that quite natural modifications of the endogenous growth models can take account of convergence, for example by introducing a gradual technological transmission between countries into the framework.

The convergence between countries observed in the real world does not deliver a decisive answer to what kind of growth model does the best job.

Growth rates and investment rates

The single empirical regularity that speaks most strongly in favour of models of (genuinely) endogenous growth is the one illustrated in Fig. 8.4: the clear positive empirical relationship between investment rates and growth rates. The endogenous growth model predicts that a higher savings or investment rate should give a higher growth rate, and indeed high investment rates do go hand in hand with high growth rates.

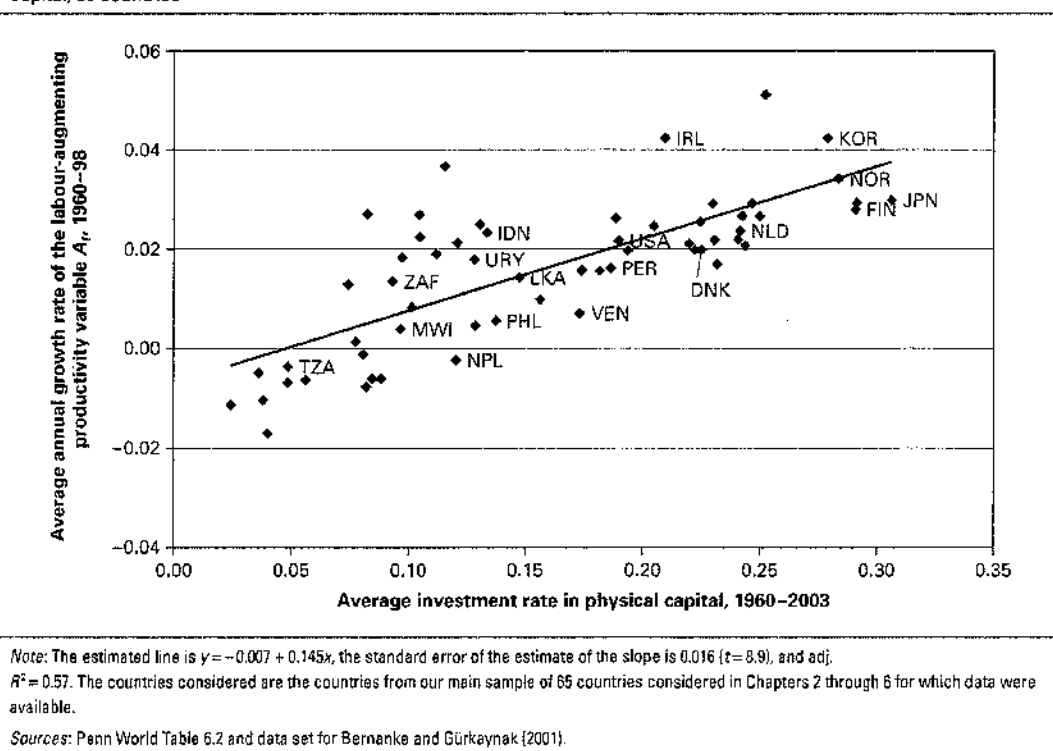
In the models of exogenous or semi-endogenous growth a higher savings rate does not give a higher growth rate in the long run, but it does so in the intermediate run due to transitory growth. The positive association between investment rates and growth rates could thus be a result of transitory growth, and therefore it does not necessarily contradict the exogenous or semi-endogenous growth models.

However, in models of exogenous growth a higher savings or investment rate, s , could not possibly give a higher growth rate in the technological variable, A_t , exactly because this growth rate is exogenous. In the endogenous growth model a higher s will give a higher growth rate in A_t , since the growth rate of A_t is $sA - \delta$, and in the semi-endogenous growth model a higher s gives a higher *transitory* growth in A_t . It is thus of interest to find out whether investment rates are also positively related to growth rates in A_t . This is indeed an idea pursued in an article by the two economists Ben S. Bernanke and Refet S. Gürkaynak.⁸ One can get an estimate of the average annual growth rate in the productivity variable A_t by growth accounting. In their article, Bernanke and Gürkaynak first do growth accounting for each of a large number of countries to get estimates of average annual growth rates of A_t . They then plot these rates against average investment rates across countries. Using the data of Bernanke and Gürkaynak the picture in Fig. 8.5 emerges. There is clearly a positive relationship. What is shown in Fig. 8.5 is the strongest empirical argument in favour of endogenous, or perhaps semi-endogenous, growth models that we know of.

The debate from which we have tried to present a few arguments is not settled. For what it is worth, the authors of this textbook find that the empirical results linking investment rates to growth rates are sufficiently convincing to make the endogenous growth research programme, and the existing models, highly promising. We also think that a serious challenge to endogenous growth theory, as it stands, comes from the empirical relationship between population growth rates and economic growth rates, which seems, if anything, to be a negative one. In contrast, semi-endogenous growth models predict a positive effect of population growth on economic growth, while endogenous growth models even predict that a constant positive population growth rate should give an exploding growth rate in GDP per capita (or that labour input is not productive at the aggregate level).

8. Ben S. Bernanke and Refet S. Gürkaynak, 'Is Growth Exogenous? Taking Mankiw, Romer, and Weil Seriously', NBER Working paper 8365, July 2001. And yes, this is the Bernanke who is now by 2010 the chairman of the US central bank, the so-called Fed.

FIGURE 8.5 Average annual growth rate of labour-augmenting productivity against average investment rate in physical capital, 60 countries



8.5 Summary

1. In endogenous growth theory the growth rate of technology, which underlies the long-run growth in GDP per worker, is endogenous: it depends on basic behavioural model parameters and therefore on structural policies that affect these parameters.
2. According to the replication argument, the individual firm's production function should exhibit constant returns to the inputs of capital and labour at any given technological level. If there are productive externalities from the aggregate stock of capital (or from aggregate production) to labour productivity or total factor productivity in the individual firm, then there can be constant returns to capital and labour at the firm level and increasing returns to capital and labour at the aggregate level. With productive externalities one can therefore maintain the convenient assumption of perfect competition and the associated theory of the income distribution, and at the same time have a potential source of endogenous growth in the model.
3. There are theoretical as well as empirical motivations for productive externalities and aggregate increasing returns. The theoretical arguments are associated with learning by doing: by being involved with (new) capital, or with production in general, workers obtain better skills. Empirical motivations come, for instance, from attempts at estimating aggregate Cobb–Douglas production functions where the sum of exponents is often found to be considerably above one.
4. Building on the idea of productive externalities, we formulated in this chapter a base model with the following features:

- The representative firm was assumed to produce output (GDP) from the inputs of labour and capital according to a usual Cobb–Douglas production function with constant returns to capital and labour and a labour-augmenting productivity variable A_t .
 - The labour productivity variable of the representative firm depended on the aggregate stock of capital as given by a function, $A_t = K_t^\phi$, $\phi \geq 0$. Since the firm should be thought of as small relative to the entire economy, it took K_t , and hence A_t , as given.
 - Capital accumulated from savings the usual way, and the (gross) savings or investment rate was a given constant. The labour force was assumed to grow at a given rate, n .
 - The real prices of capital and labour were determined by the marginal products, computed for a given level of the productivity variable.
5. It makes a qualitative difference in this model if the strength of the external effects is below, equal to, or above 1: $\phi < 1$ leads to semi-endogenous growth, $\phi = 1$ leads to (truly) endogenous growth, and $\phi > 1$ (a case we did not consider in the chapter, but which is considered in an exercise) leads to explosive growth and basically an ill-behaved model.
 6. In the case $\phi < 1$, the model implies convergence of capital per effective worker and of output per effective worker to well-defined steady state levels. In steady state there was a common constant growth rate of output per worker and of technology. This growth rate depended in a specific way on ϕ and n , and it was strictly positive if and only if both of ϕ and n were strictly positive. The feature that growth in the labour force is required for economic growth motivates the term ‘semi’ in semi-endogenous growth. The intuition is that the increasing returns driving the (semi-)endogenous growth can only be exploited if some input increases by itself. Given that the labour force grows, the capital stock will grow by a rate higher than n .
 7. For the policy implications of the model of semi-endogenous growth it is important that both the level of, and the growth rate along, the growth path for output (and consumption) per worker depend on behavioural model parameters. The levels of these paths are positively influenced by the rate of saving and investment and negatively influenced by the population growth rate. This speaks for policies to raise investment rates and dampen population growth. On the other hand, the growth rate along the steady state growth path was positively influenced by population growth, speaking for policies to promote population growth, since in the long run the positive growth effect will outweigh the negative level effect.
 8. Empirically it seems hard to find a clear positive association between population growth and growth in income per capita, although there are both pros and cons in the debate over whether growth in the labour force really promotes growth in output per worker. Given that this issue is unsettled, one would be cautious about recommending policies to promote population growth, since it seems relatively certain that higher population growth will erode the stock of capital per worker and increase the pressure on scarce natural resources, thereby lowering the level of the basic growth path, whereas the positive effect on the growth rate along this path seems doubtful.
 9. The negative association between population growth and economic growth found in the data does not contradict the model of semi-endogenous growth *per se*. If convergence to steady state is very slow, the negative level effect of higher population growth will dominate the positive growth effect for a long time. In this case, however, the steady state itself will not be very descriptive of the economy, rather the convergence process will be the relevant thing to study. The empirical evidence therefore suggests looking at the model with very slow convergence, which obtains as ϕ tends to 1.

10. We took this idea to the extreme and assumed $\phi = 1$, at the same time assuming a constant labour force. This gave us the so-called AK model. This model has constant returns to the reproducible factor, capital. It therefore implies a common and constant positive growth rate of capital per worker and output per worker, without any assumption of exogenous technological progress and without growth in the labour force being required. This is why the model's growth is called 'truly' endogenous. The everlasting balanced growth described by the model can be seen as approximating the very long-lasting transitory growth that would result if ϕ were smaller than, but close to 1. Importantly, the endogenous growth rate depended positively on the rate of saving and investment.
11. The main policy implication of the AK model was therefore that policies to promote saving and investment become even more attractive, now not only because of their level effects, but also for their permanent (or very long-lasting) effect on economic growth.
12. Empirically one finds across countries a rather strong positive relationship between, on the one hand, investment rates and, on the other hand, long-run growth rates in output per worker and in estimates of the productivity variable A_t . This is in very nice accordance with models of (truly) endogenous growth. On the other hand these models are plagued by an empirically implausible scale effect: according to endogenous growth models, a larger labour force gives higher economic growth and a growing labour force gives explosive economic growth. These features are highly counterfactual.
13. We closed this chapter by discussing the arguments for and against the theories of exogenous, semi-endogenous and endogenous growth. The scale effect implied by truly endogenous growth models speak against these models and in favour of models of semi-endogenous growth. Compared to endogenous growth models, models of exogenous growth seem easier to reconcile with the empirical evidence on convergence and on the relationship between population growth and economic growth. Nevertheless, endogenous growth models have made an interesting and promising contribution to the theory of economic growth, because of their ability to account for the observed positive relationship between investment rates and productivity growth, and because they have forced growth theorists to think harder about the forces underlying technological change.

8.6 Exercises

Exercise 1. The strength of productive externalities and the speed of convergence: simulations

This exercise illustrates by *numerical simulations* how the speed of convergence in the model of semi-endogenous growth ($0 < \phi < 1$) depends on the size of ϕ , and that for ϕ tending to 1 (here with $n = 0$), convergence becomes very slow so that the model tends to exhibit endogenous growth.

We consider the model consisting of Eqs (6)–(9), but we set $n = 0$ and we let the constant size of the labour force be $L = 1$. Hence we have excluded long-run growth due to population growth from the beginning.

1. Show that the steady state values, K^* and Y^* , of capital and output (also per worker), respectively, are:

$$K^* = \left(\frac{s}{\delta}\right)^{\frac{1}{(1-\alpha)(1-\phi)}} \quad \text{and} \quad Y^* = \left(\frac{s}{\delta}\right)^{\frac{\alpha+\phi(1-\alpha)}{(1-\alpha)(1-\phi)}}.$$

What is the steady state value of A_t ?

In the following let $\alpha = \frac{1}{3}$ and $s = 0.24$ everywhere. Let $\delta = 0.08$ initially and after a shift, fall permanently down to $\delta = 0.06$. Consider three different values for ϕ : 0.5, 0.75 and 0.9.

2. For each value of ϕ , compute the steady state values for K_t and Y_t before and after the shift in δ .
3. For each value of ϕ , let the economy be initially in steady state for $\delta = 0.08$. After some periods, 10 say, δ shifts permanently down to $\delta = 0.06$. Simulate the model over the periods before the shift and 200 periods after to create series (K_t) and (Y_t) . In one diagram plot the growth rate $(Y_t - Y_{t-1})/Y_{t-1}$ (or the approximate one, $\ln Y_t - \ln Y_{t-1}$) against periods t over the simulated periods for each of the three considered values for ϕ . Comment on the figure with respect to the opening statement of this exercise.

Exercise 2. The strength of productive externalities and the speed of convergence analytically

The transition equation, (13) or (14), we arrived at in this chapter for $\phi < 1$ can be written in the generic form $\tilde{K}_{t+1} = G(\tilde{K}_t)$ just as we did in Section 5.4 for the transition equation studied there. This is just a matter of defining G appropriately. Hence, using the same kind of operations that were used in Chapter 5 to arrive at that chapter's equation (34), a linear approximation around steady state of the transition equation studied in this chapter is:

$$\ln \tilde{Y}_{t+1} - \ln \tilde{Y}_t = \lambda (\ln \tilde{Y}^* - \ln \tilde{Y}_t), \text{ where } \lambda = 1 - G'(\tilde{K}^*).$$

1. Go through all the operations that lead to this equation yourself. (Of course, everything is the same as in Chapter 5 since there is no difference in the generic equation studied. It is important, though, that you can do the full linear approximation yourself.) Explain in what sense λ is a measure of the rate of convergence for $\tilde{Y}_t$.
2. Show that the rate of convergence (for $\phi < 1$) is:

$$\begin{aligned} \lambda &= (1 - \alpha)(1 - \phi) \frac{(1 + n)^{\frac{1}{1-\phi}} - (1 - \delta)}{(1 + n)^{\frac{1}{1-\phi}}} \\ &= (1 - \alpha)(1 - \phi) \left[1 - \frac{1 - \delta}{(1 + n)^{\frac{1}{1-\phi}}} \right]. \end{aligned}$$

Convince yourself that in the special case $\phi = 0$, this rate is in accordance with the rate found in Chapter 5. Show that as ϕ goes to 1, λ goes to 0. Try to explain intuitively why a large ϕ implies slow convergence.

Exercise 3. The model of semi-endogenous growth ($\phi < 1$) and of endogenous growth ($\phi = 1$) with the productive externality arising from K_t/L_t rather than from K_t

In our discussion in Section 8.3 of the scale effect in connection with the model of endogenous growth ($\phi = 1$) we said that one can eliminate the effect by assuming a productive externality of the form:

$$A_t = \left(\frac{K_t}{L_t} \right)^\phi,$$

rather than the form $A_t = K_t^\phi$ mainly assumed in the chapter. We now consider the model of *semi*-endogenous growth ($\phi < 1$) with the alternative form of the external effect and we study the model of endogenous growth ($\phi = 1$) a little further to show that with the alternative formulation of the external effect one can have population growth in the model without getting exploding economic growth.

The model to be considered is (6)–(9), but with (7) replaced by the equation above. Our assumptions are the same as in the chapter, among them $s > n + \delta$.

Before turning to the more technical questions, try to think about whether you find the form $A_t = K_t^\phi$ or the form $A_t = (K_t/L_t)^\phi$ most convincing for the productive externality. Should the external effect on labour productivity from the use of machinery be expected to come

from the amount of capital as such or from capital per worker? Think, for instance, of a firm with ten workers beginning to use computers. The direct (privately appropriated) productive effect will, of course, be stronger with ten computers than with one. How about the indirect (external) effect?

For questions 1–5 assume that $0 < \phi < 1$.

1. Show that now:

$$\frac{A_{t+1}}{A_t} = \frac{\left(\frac{K_{t+1}}{K_t}\right)^\phi}{(1+n)^\phi},$$

and that with the usual definitions, the transition equation is:

$$\bar{k}_{t+1} = \left(\frac{1}{1+n}\right)^{1-\phi} \bar{k}_t (s\bar{k}_t^{\alpha-1} + (1-\delta))^{1-\phi}.$$

Convergence to positive steady state values, $\bar{k}^*$ and $\bar{y}^*$, follows from the properties of the transition equation the same way as in this chapter.

2. Find the steady state values, $\bar{k}^*$ and $\bar{y}^*$, for $\bar{k}_t$ and $\bar{y}_t$ respectively. Show that in steady state the common growth rate of A_t , k_t and y_t is 0 (also when $n > 0$).
3. What is the aggregate production function in this model? Is labour unproductive at the aggregate level (also when $\phi < 1$)? Why is it that in this case there is no positive, semi-endogenous type of growth in GDP per worker in steady state?

The fact that the presence of the external effects does not produce (semi-)endogenous growth in the steady state this time may give the impression that ϕ is not economically important, at least not for steady state. The size of ϕ is, however, important both for the steady state levels and for the convergence process outside steady state, as the answers to the following two questions should reveal.

4. Show that the constant levels of capital per worker and output per worker in steady state are:

$$k^* = \left(\frac{s}{n+\delta}\right)^{\frac{1}{(1-\alpha)(1-\phi)}} \quad \text{and} \quad y^* = \left(\frac{s}{n+\delta}\right)^{\frac{\alpha+\phi(1-\alpha)}{(1-\alpha)(1-\phi)}}.$$

5. By considering again the linear approximation around steady state, etc., find, also for this model, the rate of convergence λ for $\bar{y}_t$ according to the linear approximation. Show that (again) λ goes to 0 as ϕ goes to 1. (The latter result means that the endogenous growth model ($\phi = 1$) without the scale effect approximates the behaviour of the model considered here for large ϕ .)

Now assume $\phi = 1$, and still $n > 0$.

6. Show that the Solow equation is:

$$k_{t+1} - k_t = \frac{1}{1+n} (s - \delta - n)k_t,$$

and that the endogenous growth rate of GDP per worker is $g_e = (s - \delta - n)/(1 + n)$. Comment with respect to scale effects.

Exercise 4. Balanced growth

Show that the steady state growth path of the model of semi-endogenous growth considered in Section 8.2 is one of balanced growth. Then show that the growth path of the model of endogenous growth considered in Section 8.3 (with $n = 0$) is one of balanced growth. (Essentially you are asked to show that all of capital per worker, output per worker and wages per worker (the wage rate) grow by one and the same constant rate and that the real

rental (interest) rate of capital is constant along both of these growth paths. Since the models assume a constant growth rate of the labour force (possibly 0) the other requirements for balanced growth follow.)

Exercise 5. Explosive endogenous growth

You may have been wondering about some cases not dealt with in this chapter. For instance, when we were setting $\phi = 1$ in Section 8.3 (as an approximation of a large ϕ just below 1) we assumed at the same time $n = 0$. This was perhaps for a good reason, but $\phi = 1$ and $n > 0$ have not been considered. Furthermore, we have considered $\phi < 1$ and $\phi = 1$, but not $\phi > 1$. This exercise asks you to demonstrate that the uncovered cases lead, or may lead, to explosive growth, that is, to a growth rate in GDP per worker that increases over time and eventually goes to infinity. This is, of course, a very unrealistic feature.

Consider the growth model consisting of the equations (6)–(9).

1. In that model, let $\phi = 1$ and $n > 0$ (so that $L_t = (1 + n)^t L_0$ increases at a constant rate n). Show that the (exact) growth rates of k_t and y_t , respectively, in period t are:

$$\frac{k_{t+1} - k_t}{k_t} = \frac{1}{1+n} [sL_0^{1-\alpha}(1+n)^{(1-\alpha)t} - (n+\delta)]$$

and

$$\frac{y_{t+1} - y_t}{y_t} = \frac{1}{(1+n)^\alpha} [sL_0^{1-\alpha}(1+n)^{(1-\alpha)t} + 1 - \delta] - 1.$$

2. Now let $\phi > 1$ (but say also $\phi < 2$, not to be extreme) and $n = 0$ (so that L_t is constant, L say). Show that the aggregate production function can be written $Y_t = AK_t^{1+(1-\alpha)(\phi-1)}$, where $A \equiv L^{1-\alpha}$ (and since $\phi > 1$, the exponent on K_t is greater than 1). Write down the transition equation for K_t and show that K_t has a unique steady state value: $K^* = [\delta/(sA)]^{1/[(1-\alpha)(\phi-1)]}$. Is anything puzzling you about this value? Write down the Solow equation for K_t and sketch the Solow diagram. Is K^* stable or unstable? Also state the modified Solow equation and draw the modified Solow diagram. If K_t is initially greater than K^* , how does the growth rate of K_t evolve over time? How about the growth rates of k_t and y_t ? What if $K_t < K^*$ initially?

Exercise 6. Endogenous growth with both physical and human capital

In this chapter we presented endogenous growth models with physical capital, but without human capital. The following three equations make up a model of endogenous growth including human capital:

$$Y_t = AK_t^\nu H_t^{1-\nu}, \quad 0 < \nu < 1.$$

$$K_{t+1} = s_K Y_t + (1-\delta)K_t$$

$$H_{t+1} = s_H Y_t + (1-\delta)H_t.$$

Notation and parameter values are as usual if not explicitly stated. The first equation is the aggregate production function where the influence of the constant labour force, L , has been put into A and the sum of the exponents on physical and human capital add up to 1, so there is not diminishing, but constant returns to the reproducible factors taken together. The next two equations describe the accumulation of physical and human capital, respectively, just as in Chapter 6.

1. First show that if the production function of the individual firm is:

$$Y_t = (K_t^d)^\alpha (H_t^d)^\phi (A_t L_t^d)^{1-\alpha-\phi},$$

and an external productive effect implies:

$$A_t = (K_t^{\frac{\alpha}{\alpha+\phi}} H_t^{\frac{\varphi}{\alpha+\phi}})^{\alpha+\phi},$$

then if $\phi = 1$, the aggregate production function above results, where $A = L^{1-\alpha-\varphi}$, and $\nu = \alpha/(\alpha + \varphi)$. How should we view this model bearing in mind that a knife-edge case such as $\phi = 1$ is not of interest in itself?

2. Define the ratio between physical and human capital, $x_t \equiv K_t/H_t$. Show that the model above implies the following dynamic equation for this ratio:

$$x_{t+1} = \frac{s_K A x_t^\nu + (1-\delta)x_t}{s_H A x_t^\nu + (1-\delta)},$$

and demonstrate that this equation has properties that imply that in the long run x_t converges to $x^* = s_K/s_H$.

3. Show that according to the model above the growth rates of physical and human capital, respectively, in any period t are:

$$g_t^K \equiv \frac{K_{t+1} - K_t}{K_t} = s_K A \left(\frac{K_t}{H_t} \right)^{\nu-1} - \delta,$$

$$g_t^H \equiv \frac{H_{t+1} - H_t}{H_t} = s_H A \left(\frac{K_t}{H_t} \right)^\nu - \delta,$$

and then show that as $x_t = K_t/H_t$ tends to its long-run equilibrium value x^* , these two growth rates tend to the common value:

$$g = s_K^\nu s_H^{1-\nu} A - \delta.$$

Show that in long-run equilibrium g is also the constant and endogenous growth rate of Y_t and of y_t .

4. According to this model, growth in GDP per worker depends positively not only on s_K as tested in Fig. 8.4, but also on s_H . To test this, create a diagram that plots growth rates in GDP per worker, g^i , against investment rates in human capital, s_H^i , across countries i with data taken from Table A (includes countries from the sample of 65 countries). Comment.

From the empirical observations we described in Chapter 6, α and φ should be of about equal size, which should here imply $\nu = \frac{1}{2}$. Put the model's full (endogenous) growth prediction to a test by plotting growth rates g_i against $(s_K^i)^{1/2}(s_H^i)^{1/2}$ again taking data from Table A. Estimate the line of best fit. Compare with Fig. 8.4 and to the figure plotting g^i against s_H^i . Comment.

Exercise 7. Semi-endogenous growth and endogenous growth if the productive externality arises from Y_t rather than from K_t

As mentioned in this chapter, sometimes models with productive externalities assume that the external learning-by-doing effect arises from workers being involved in production as such rather than (more narrowly) from their being involved with the use of capital.

We therefore consider a growth model consisting of the four equations (6), (8), (9) and:

$$A_t = Y_t^\phi, \quad 0 < \phi < \frac{1}{1-\alpha},$$

where the latter equation replaces (7).

1. Show that the aggregate production function in this model is:

$$Y_t = K_t^{\frac{\alpha}{1-\phi(1-\alpha)}} L_t^{\frac{1-\alpha}{1-\phi(1-\alpha)}}.$$

Why have we assumed $\phi < 1/(1-\alpha)$? Show that the aggregate production function has increasing returns to (K_t, L_t) whenever $\phi > 0$. Show also that if $\phi = 1$ it has constant returns to K_t alone.

2. Show that in this model:

$$\frac{A_{t+1}}{A_t} = \left(\frac{K_{t+1}}{K_t} \right)^{\frac{\alpha\phi}{1-\phi(1-\alpha)}} \left(\frac{L_{t+1}}{L_t} \right)^{\frac{\phi(1-\alpha)}{1-\phi(1-\alpha)}}$$

Assume now that $\phi < 1$. Let $\tilde{y}_t$ and $\tilde{k}_t$ be defined as usual: $\tilde{k}_t \equiv k_t/A_t \equiv K_t/(A_t L_t)$ and $\tilde{y}_t \equiv y_t/A_t \equiv Y_t/(A_t L_t)$.

3. Show that the transition equation for $\tilde{k}_t$ is:

$$\begin{aligned} \tilde{k}_{t+1} &= \left(\frac{1}{1+n} \right)^{\frac{1}{1-\phi(1-\alpha)}} \tilde{k}_t [s\tilde{k}_t^{\alpha-1} + (1-\delta)]^{\frac{1-\phi}{1-\phi(1-\alpha)}} \\ &= \left(\frac{1}{1+n} \right)^{\frac{1}{1-\phi(1-\alpha)}} (s\tilde{k}_t^{\frac{\alpha}{1-\phi}} + (1-\delta)\tilde{k}_t^{\frac{1-\phi(1-\alpha)}{1-\phi}})^{\frac{1-\phi}{1-\phi(1-\alpha)}} \end{aligned}$$

Find the steady state values for $\tilde{k}_t$ and $\tilde{y}_t$, and show that these are meaningful whenever $(1+n)^{1/(1-\phi)} > (1-\delta)$ (which we assume). Show also that the transition equation implies convergence to steady state.

4. Find the expression for the growth rate, g_{ss} , of output per worker in steady state. Comment with respect to what creates growth in output per worker in this model (when $\phi < 1$).

Now assume $\phi = 1$, and $n = 0$, so L_t is equal to some L for all t .

5. Show that the model can be condensed to the two equations:

$$Y_t = AK_t, \quad A \equiv L^{\frac{1-\alpha}{\alpha}},$$

$$K_{t+1} = sY_t + (1-\delta)K_t.$$

Find the growth rate, g_n , of output per worker. Comment, for instance, with respect to scale effects.

Exercise 8. Taxation and productive government spending: endogenous growth without productive externalities

There is a simple alternative to productive externalities as a mechanism for endogenous growth, which yields basically the same models as studied in this chapter: productive government expenditure financed by (proportional) income taxation. The model studied in this exercise comes from Barro (1990),⁹ and consists of the five equations below.

The production function is as usual with a labour productivity variable A_t :

$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}.$$

Income is taxed proportionally by the rate τ , $0 < \tau < 1$, in all periods, so the tax revenue in period t is τY_t . The budget is balanced, so government expenditure in period t is:

9. Robert J. Barro, 'Government Spending in a Simple Model of Endogenous Growth', *Journal of Political Economy*, 98, 1990, pp. S103–S125.

$$G_t = \tau Y_t.$$

Government spending (on infrastructure, the legal system, health care, etc.) is assumed to be 'productive' in the sense that productivity is influenced positively by expenditure:

$$A_t = aG_t, \quad a > 0.$$

Total savings and investment are given as a given fraction of total *net* income, so the capital accumulation equation is:

$$K_{t+1} = s(1 - \tau)Y_t + (1 - \delta)K_t.$$

Finally we assume a constant labour force:

$$L_t = L.$$

1. Show that the aggregate production function (where it is taken into account how A_t depends on Y_t through taxation, etc.) is:

$$Y_t = (a\tau L)^{\frac{1-\alpha}{\alpha}} K_t,$$

and comment.

2. Show that the model is equivalent to an AK model (like (20) and (21) in the chapter) with appropriate relabelling of parameters. Find the growth rate of capital per worker and of GDP per worker that the model gives rise to.
3. Is there a scale effect? Assume alternatively that it is government expenditure per worker, G_t/L_t , that is productive, so that $A_t = aG_t$ above is replaced by $A_t = a(G_t/L_t)$. Analyse the model and find the growth rate of GDP per worker. Comment with respect to the scale effect and the aggregate productivity of labour.
4. What value of τ maximizes the rate of economic growth (for the case $A_t = aG_t$ as well as for $A_t = a(G_t/L_t)$)? Describe the opposing forces that determine the growth-maximizing tax rate. Discuss the policy implications of your analysis. How would policy implications be affected, do you think, if there were traditional distortionary effects of taxation, e.g., if labour supply was an increasing function of the after-tax real wage?

Consider a generalization of the model where the equation $A_t = aG_t$ is replaced by $A_t = a(G_t)^\phi$, $0 < \phi \leq 1$, and $L_t = L$ is replaced by $L_{t+1} = (1 + n)L_t$. The model above corresponds to $\phi = 1$, and $n = 0$, so let us now assume $\phi < 1$ and $n > 0$.

5. Analyse the resulting model as we have analysed models of semi-endogenous growth in this chapter, going through the following steps: find the aggregate production function and describe its returns to scale. Find the transition equation for $\tilde{k}_t \equiv k_t/A_t \equiv K_t/(A_t L_t)$, verify convergence to a steady state, and find the growth rate of output per worker and of A_t in steady state. Comment with respect to policy implications. (Hint: Exercise 7 can be of help here.)

Exercise 9. Exogenous or endogenous growth models?

You have now seen some models of exogenous growth and some of endogenous growth and you have been presented with some pros and cons for both types of models. At this stage, what kind of model do you think gives the best understanding of economic growth? This is not the type of exercise that has one correct answer. In fact, leading growth economists still discuss this issue between them. Get together with fellow students and line up the most important pros and cons for each type of model. Then discuss which arguments you find most important and convincing, and finally try to make up your own mind on the issue: exogenous or endogenous growth.